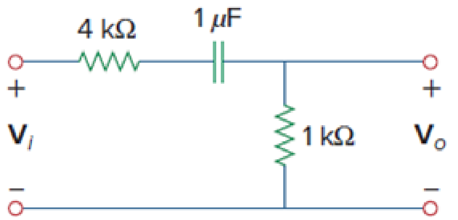
# Chapter 14, Problem 84P

## Problem

Using *PSpice or MultiSim,* obtain the frequency response of the circuit in Fig. 14.101 on the next page.

**Figure 14.101:**




## Step-by-step solution

**Step 1** of 4

Open the PSpice and click on draw tool then select place part and type R and choose the resistors of value $4\ \text{k}\Omega$ and $1\ \text{k}\Omega$.

Next type the letter C and choose the capacitor of value $1\ \mu\text{F}$.

Type $\text{V}_{\text{ac}}$ and $\text{agnd}$ to select the ac voltage source of amplitude $1\ \text{V}$ and ground.

Connect all the parts using wire to draw the schematic as shown in Figure 1.

Insert a voltage marker at the output to measure output voltage at resistor $R_2$.

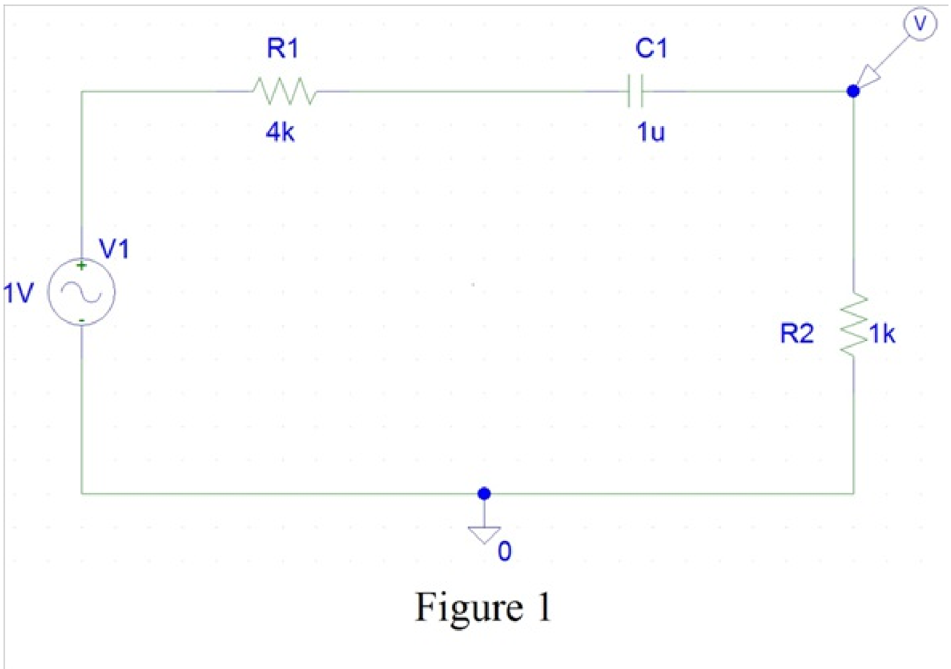



Figure 1

**Step 2** of 4

To determine the frequency reponse simulate the circuit shown in Figure1.

Click on setup analysis and enable AC sweep.

Then AC sweep dialogue box will appear choose decade in AC sweep type.

Select $1\,\text{Hz}$ as start frequency and $1\,\text{kHz}$ as end frequency.

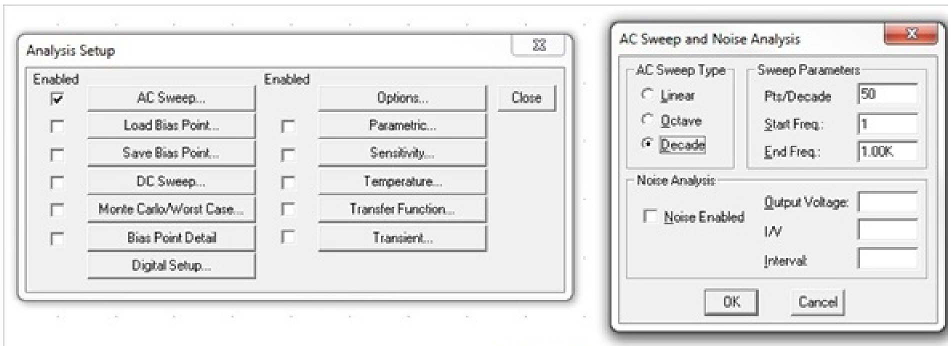



Figure 2

## Step 3 of 4

Simulate the schematic and obtain the results.

The magnitude plot of frequency response is shown in Figure 3.

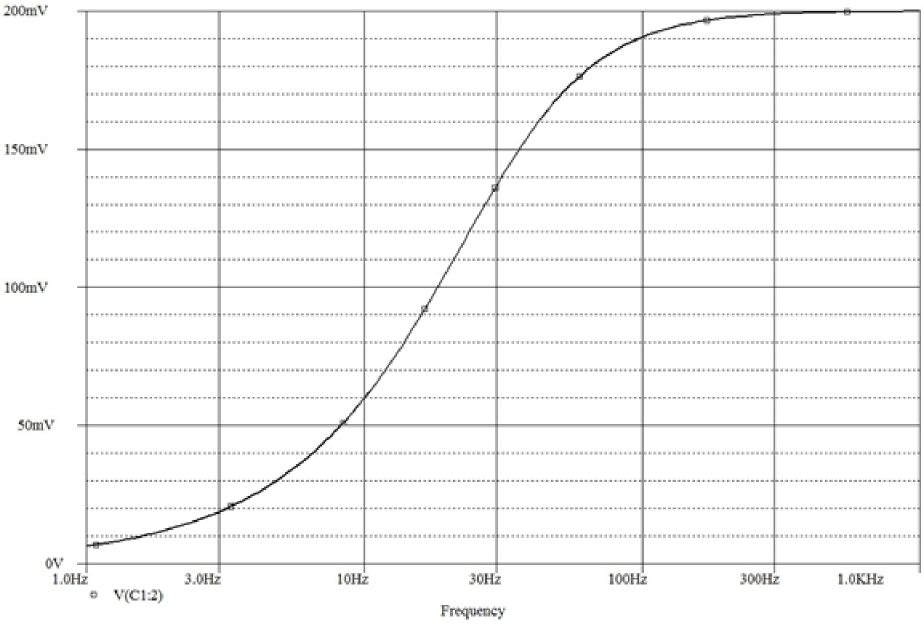



Figure 3

## Step 4 of 4

To determine the phase response click on add trace and write the expression $P(VR2)$.

Then the phase plot is obtained as shown in Figure 4.

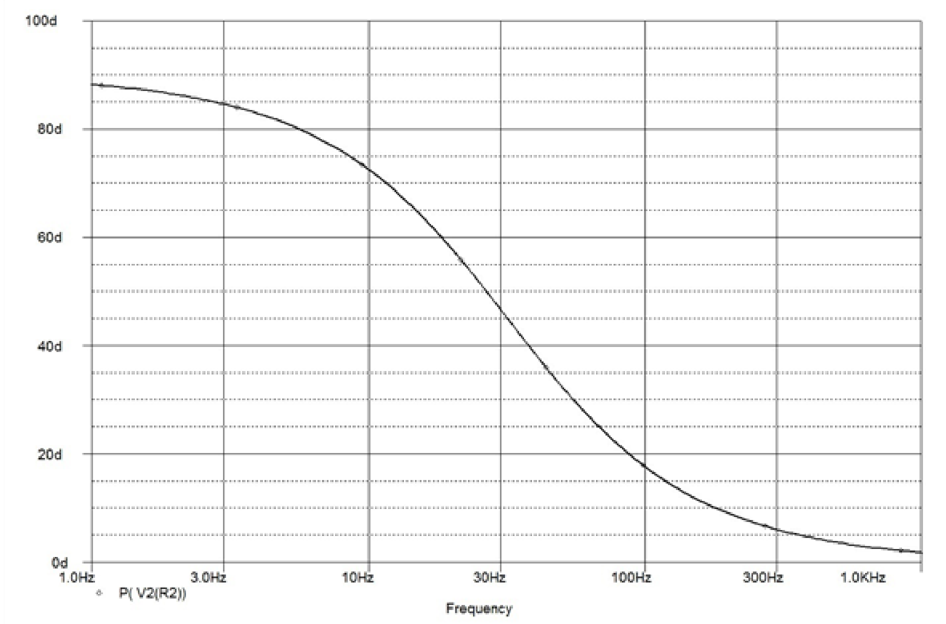



Figure 4